Akin **Eker**

Blockchain Research Engineer, PhD in Computer Science
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Summary

Blockchain Engineer and Research Associate specialised in consensus protocols, scalable blockchain systems and zk-rollups, with 4+ years' experience designing and building scalable blockchain architectures.

Core: Consensus protocol research (taxonomy of 37 protocols), ZK-rollup system architectures (L1 contracts + Rust services), Applied cryptography, Distributed systems, L1/L2 trade-offs, Single-chain and multi-chain systems.

Skills

Coding: Solidity | Rust | C++ | Python | JavaScript | SQL

Blockchain: Consensus Protocols | ZK-Rollup Architectures | Merkle Trees (Poseidon/Keccak) | EVM OpCodes and Precompiles

Development Tools & Frameworks: Git | Docker | snarkjs | Foundry | Hardhat | Truffle | Node.js | Chainlink | OpenZeppelin | AWS | Linux

Simulation & Protocol Analysis: Consensus Protocols | Agent-Based Simulation | Game Theory

Experience

Research Associate - ELABORATOR Project
Bristol, UK
University of Bristol — *2026 – Present*

Collaborating within an international research team on the ELABORATOR project to investigate blockchain and privacy-preserving frameworks.

PhD Researcher
Bristol, UK
University of Bristol — *2021 – 2026*

- **EduRoll: Ethereum Layer 2 Zero-Knowledge Rollup Implementation *(Solidity, Rust, Circom, snarkjs, Foundry, PostgreSQL, Docker)***
  - Designed and implemented a modular zk-rollup prototype for education-related payments on Ethereum (research project), demonstrating the full execution, proving, and on-chain verification pipeline for batched L2 token transfers.
  - Designed a hybrid hash architecture that decouples in-circuit prover cost from EVM compatibility: a fixed-depth (depth-20) Poseidon Merkle tree for L2 state and Keccak-256 for L1 commitments.
  - Selected Groth16 over PLONK/Halo2/STARK families after weighing on-chain verifier cost against trusted-setup trade-offs for an EVM-targeted rollup.
  - Implemented the transfer circuit in Circom and integrated snarkjs Groth16 proving in the off-chain prover.
  - Developed the L1 Solidity contracts (Rollup, auto-generated Groth16 verifier, ERC-20 bridge) that verify each batch's proof on-chain and post the batch's transactions as calldata for data availability.
  - Built an asynchronous off-chain pipeline (Rust + PostgreSQL) in which the Sequencer, Prover, Submitter, and Archiver coordinate through a persistent database rather than direct IPC, providing fault tolerance and clean restart semantics under partial failure.
  - Deployed and benchmarked the system on a local Anvil devnet: 20 transactions per batch ($\sim$628k non-linear circuit constraints ($\sim$1.17m in total)), with on-chain batch submission (Groth16 verification plus calldata DA) at $\sim$327k gas, i.e. $\sim$16k gas amortised per transfer.
- **FlexBoT: A Scalable IoT Architecture Proposal and Simulations *(Solidity, C++, Python, Docker, ContikiNG IoT Simulator)***
  - Designed and implemented a scalable blockchain framework for IoT device coordination and multi-application management.
  - Implemented Solidity smart contracts for runtime application switching and secure firmware updates.
  - Developed C++ firmware modules for IoT clusters which runs in containerised Docker environment.
  - Implemented Python-based data post-processing and visualisation for system evaluation.
  - Simulated performance under varying network loads and network sizes, demonstrating a scalable, low-latency architecture (*$O(\log n)$ complexity*).
- **Consensus Protocol Research**
  - Analysed and classified 37 consensus protocols to evaluate trade-offs across security, scalability, and decentralisation (the Blockchain Trilemma).
  - Identified algorithmic design principles that enable both application-aware and application-agnostic blockchain systems, optimising system performance for specific needs/constraints.

Teaching Assistant
Bristol, UK
University of Bristol — *2022 – 2025*

- **Ethereum - FinTech Group Projects:** Assisted in smart contract development and testing, project management, and technical reporting.
- Assisted labs, marked assessments, and supervised individual and group projects.

Research and Teaching Assistant
Ankara, Turkey
Gazi University — *2016–2021*

- Delivered and assessed units and supervised student projects.
- **Units:** Data Structures and Algorithms | Database Management Systems | Artificial Intelligence | Machine Learning

Education

University of Bristol *Bristol, UK*
PhD in Computer Science *2021 - 2026*
- Thesis: *Blockchain-of-Things: Novel System Architectures and Consensus Models as a Scalable Architecture for IoT-Driven Smart Environments*

University of Southampton *Southampton, UK*
Master of Science in Computer Science *2018 - 2019*
- Thesis: *Recognising and Detecting Activities in ADL Datasets using Artificial Intelligence*

Gazi University *Ankara, Turkey*
B.S. in Computer Engineering *2010 - 2015*
- Final Project: *Read a Book: A Web Platform that Allows Reading, Purchasing Books, and Note-taking and Researching on One Platform*

Selected Publications & Technical Research

2026 **Deconstructing Consensus: A Taxonomy of Protocols, Blockchain Trilemma, System Properties, and Democratic Design**, *Under Review*

2026 **Blockchain-of-Things (BoT): A Layered Architecture and Performance Evaluation for Scalable IoT Environments**, *Under Review*

2024 **FlexBoT: A Scalable Architecture for Multi-Application Supporting BoT Environments with Application Shifting at Runtime**, IFIP Networking Conference *Greece*

Selected Awards & Certificates

Certificates

2022 **Innovation and Enterprise Certificate**, University of Bristol *Bristol, UK*
2015 **Seeds for the Future Programme Certificate**, Huawei Inc *Shenzhen & Beijing, China*

Awards

2022 **Bristol Plus Award**, University of Bristol *Bristol, UK*
2016 **Turkish Ministry of Education Scholarship**, Postgraduate Scholarships Programme *Ankara, Turkey*
2012 **Exchange Student, University of Vigo**, Erasmus Programme in Computer Science *Ourense, Spain*

Program Committees

2026 **Reading Panellist**, SparckAI Programme, University of Bristol *UK*
2026 **Peer Reviewer**, IEEE International Conference on Cyber Security and Resilience *Lisbon, Portugal*
2025 **Peer Reviewer**, IEEE International Conference on Cyber Security and Resilience *Crete, Greece*
2024 **Peer Reviewer**, Ad-Hoc Networks Journal, Elsevier *Bristol, UK*
2023 **Peer Reviewer**, IEEE International Conference on Communications *Rome, Italy*